Self-Surveying Multi-Robot System for Remote Deposition Modelling

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The need for increased automation of the AEC sector has been extensively documented within the architectural discipline over recent years. Far beyond economic perspectives, current advances in technology offer an increased and more direct implementation of sustainable materials. Within this research, the potential for the re-use of material with low embodied energy within automated construction will be examined. Herefore, Remote Material Deposition (RDM, firstly described in Dörfler et al., 2014) is utilized as main fabrication method, deploying varying compositions of local building debris, lime mortar, and sand, via a throwing arm. This research explores a method of continuous verification of material deployment and removal of material oversaturation to guarantee accuracy. Herefore, all instances of the robot ecology are in direct communication with one another and the user for verification, adaptation, and information.
The proposed framework is examined through experimentation by designing, building, and implementing an inter-communicative network of bespoke semi-autonomous robots with all proposed parts of the system.

Keywords: *Construction automation, Material reuse, Onsite construction, Self-verifying system, Robot ecology, Additive manufacturing*

INTRODUCTION

Construction and demolition waste (C&DW) amounts reportedly to 374 million tons in the EU-28 in 2016 alone (European Environment Agency, 2019). A further, steady rise in the total mass of C&DW can be expected (European Commission, 2011). While through the establishment of markets for the recovery of C&DW, the target set by the 2008 Waste Framework directive is either archived or surpassed by most EU members, large portions thereof are fulfilled by backfilling of these materials, and therethrough eroding its inherent material value. (European Environment Agency, 2019). The potential for the reuse of such low embodied energy materials as a path towards a more sustainable building industry as a circular economy enabled through scanning and sensing technologies has previously been demonstrated by a.o. Johns et al 2020, and Clifford and McGee, 2018. However, while relying on these technologies and witnessing the rise of robotic development for fabrication within the last decade, the construction industry to date remains largely under-automated (McKinsey Global Institute, 2017).

While a larger potential of automation as a support for construction is being discussed, specific tasks within the process have already been successfully tested and utilized through distributed robotic units, e.g. as explored site surveying, automated material removal and transport, or

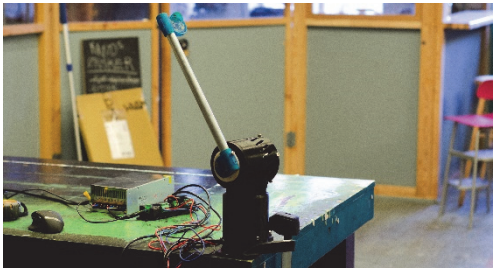
specialized selective material deposition. (Claypool, 2019)

This research explores the potential of the technology found in specialized distributed units as unified inter-communicative robot ecology. Here, continuous verification processes of material deployment allow for the reuse of C&DW materials. A key investigation is the development of a system for irregular and heterogeneous material composition, weight, and shape. Herefore, this research asks the question:

Can inter-communicative distributed units, tasked with closed-loop fabrication, allow the repurposing of Construction and Demolition Waste and aid the development of sustainable automated construction?

STATE OF THE ART

Research on automated on-site fabrication has targeted challenges posed by conventional construction sites (Giftthaler et al., 2017) through the application of mobile fabricating robots, as demonstrated a.o. by Gramazio Kohler Research Group and Agile & Dexterous Robotics Lab, largely with the In-Situ Fabricator 1, (Sandy et al., 2016). This successfully demonstrated the feasibility of closed-loop fabrication, allowing for continuous process verification.



The implementation of sensory feedback here allows for high-accuracy construction at full building scale. (Giftthaler et al., 2017).

Further, Feedback Control Systems (FCS) allow for direct adaptability of the fabrication process to the construction site (Dörstelmann, 2015).

Features like active 3-dimensional map building enable it to 'understand' its environment more directly, without the need for constant human supervision. While this, on one hand, provides opportunities over inaccessibility of the site, visual control systems further allow for additional accuracy within the building process (Lussi, 2018). Additionally, these systems can provide real-time information about the state of fabrication and challenges within the direct environment (Tish et al., 2020) and the active correction of occurring errors throughout the process (Dörfler et al., 2019), allowing the use of materials beyond highly calibrated factory conditions for site and material. This enables and eases the integration of natural materials of unknown geometry within fully automated fabrication as demonstrated by Eversmann (2018).

METHOD

The utilization of closed-loop fabrication for the application of re-used, untuned, and uncalibrated data is being investigated within this research. Herefore, the research explores the development and implementation of a semi-autonomous collaborative multi-robot setup for on-site fabrication. The task of closed-loop fabrication is discretized into three subtasks, fulfilled by corresponding members of a robot ecology.

These subtasks are examined by material experiments of individual or sets of robotic units in the following paragraphs: 'Material deposition', 'Site survey and fabrication evaluation' and 'Finishing'.

Figure 1
Early prototype of 2 DOF throwing arm

Material Deposition

The research relies on robotic throwing as a unified method of on-site material transport as well as fabrication technique. As conventional AM processes lack opportunity for scalability and verify-ability (Frank, 2008), remote material deposition (RMD), as brought forth by Dörfler et al. (2014), was able to demonstrate significant advances towards this goal. Through the use of verified robotic throwing, Dörfler et al. were able to largely

overcome conventional AM-related size constraints allowing for fabrication at full building scale.

For this reason, this research utilizes the RMD process to deploy material from a custom, stationary, two degrees-of-freedom (DOF) throwing arm (see Figure 1).

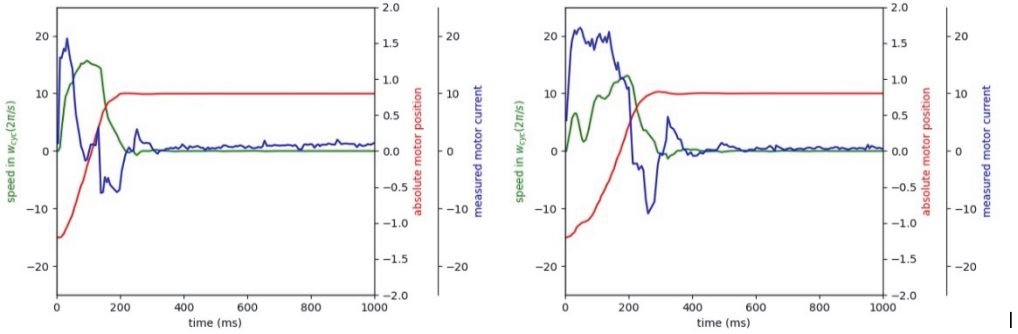


Figure 2
Measured data via the servo drive module showing robotic arm speed, motor position, and measured motor current, showing near identicall launch speeds with varying load (0g and 802g)

Both joints of the throwing arm consist of a mostly 3D-printed modular system, utilizing brushless motors, controlled by an Odrive servo drive module (Weigl, 2020) and a variation on the OpenTorque planetary gearbox (Levine, 2019) with 8:1 reduction, allowing for high flexibility through their compliance as well as high rotational velocity. This approach allows for the utilization of a compliant mechanism at a far reduced price compared to conventional comparable mechanisms (Gealy et al., 2019), positioning itself as a compromise of agility of directly driven actuators and torque efficiency of a high reduction gear train (Kau et al., 2019).

By its low reduction ratios, acting as Quasi Direct Drive system (QDD), both axes allow for rapid movement up to 8 cycles per second (ω_{cyc}) resulting in tangential launch speeds of 19m/s. The QDD reduction further allows for acceleration of the throwing axis over a shortened effective path.

With the servo drive module, it is possible to keep the throwing arms' launch velocity nearly identical throughout varying loads, allowing for approximately identical throwing trajectories for a set of tested projectiles (see Figure 2).

Variations of material density, and therefore the launched pellets' composition remains irrelevant to the fabrication process, eliminating the need for calibrated or previous knowledge of the material.

is proposed to be continuously examined by an autonomous surveying unit (see figure 03). The unit utilizes a differential drive unit, where bi-directional movement of the hub motor allows zero turn radius, therefore allowing for high maneuverability and velocity with low mechanical complexity. (Delgado-Mata, 2012). Initial site survey as well as localization enabled by Simultaneous Localization and Mapping (SLAM) through hector-SLAM (Kohlbrecher and Meyer, 2014). This allows for building a 2-dimensional map and further path finding and cost mapping tasks. Active localization provides

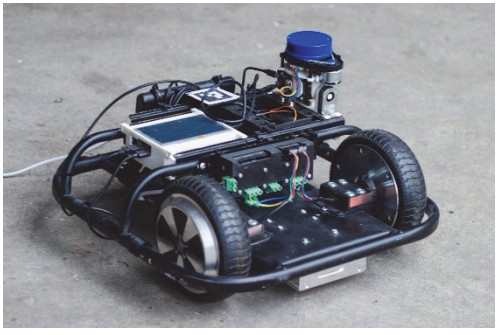


Figure 3
Surveying WMR prototype

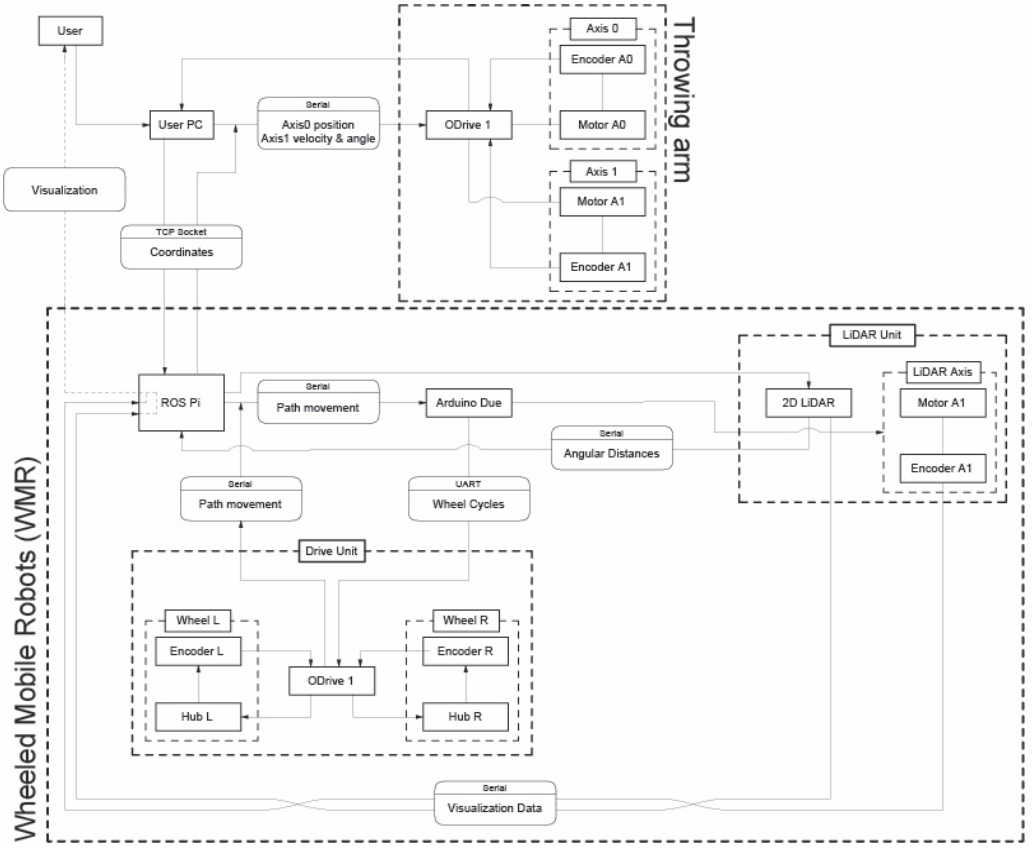


Figure 4
Closed-loop operation of surveying WMR and inter-communication with throwing arm and user, highlighting communication between robotic units (bold dashed container), and flow of data between MCUs (in bold container) , sensors and actuators

orientation and informed operation of the Wheeled Mobile Robot (Elfes, 1989) and can further build enhanced information maps for the use of the secondary units. Herefore, the WMR is equipped with a 360-degree two-dimensional LiDAR rangefinder. An additional third axis allowing for tilting of the LiDAR unit measurements in 3-dimensional space can be taken, allowing for verification of specific deployment areas. This allows specific measurements of areas in question, allowing for more calculation efficient processes through small point cloud sizes (see figure 5).

Herefore, the axis is actuated from the main units control system relayed via a microcontroller (MCU). The MCU used as a secondary control unit allows for eased data handling and software development as well as individualized testing.

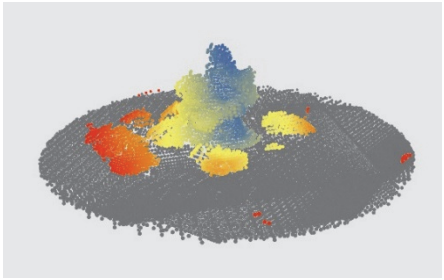


Figure 5
Point cloud sampling via surveying unit

The MCU confirmed concluded processes back to the main control unit, which then relays the information to the human user for visualization. (see process overview in figure 4)
Site and fabrication surveying are conducted through a SLAM process built on Open Robots 'Robotic Operating System' (ROS Kinetic Kane). This allows for active localization of the unit itself while conducting map-building with geometrical data, enabling the collection and analysis of additional data such as cost-mapping for trajectory planning.

Automated finishing

High surface quality and part accuracy has remained a core challenge of large scale 3d printing (Buswell et all, 2018). Hack and Kloft (2020) were able to overcome this by post-print smoothing and edge finishing. A similar process is applied here with the help of an additional finishing unit, 3-DOF robotic arm (see figure 6) on a WMR base. This allows for more precise connections to additional architectural elements (as simulated, see figure 7), as well as addressing material oversaturation in events of wrongful deployment or material rolloff. The arm itself builds on the joints previously developed for the throwing arm adding a third unit for an additional degree of freedom. The unit is proposed to sit on the platform similar to the aforementioned surveying unit, utilizing its SLAM technology for self-localization. Thereby the correct path trajectory can be verified, limiting positioning problems. Additional measuring of the 'finished' surfaces can be verified by the measuring unit.

Figure 6
Finishing arm prototype

Figure 7
Finishing process for attachment of additional architectural elements

Material

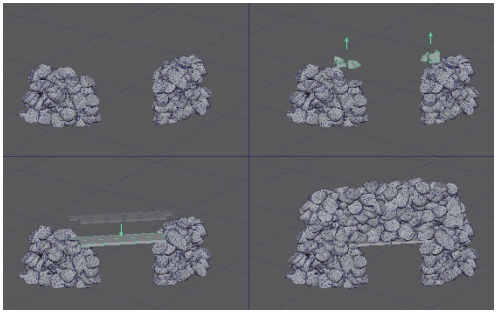
The material used within this research is modeled on the potential of reusing of C&DW with low embodied energy for the specific purpose of throwing. The composition is developed within the scope of the examination of the proposed and developed robot ecology and has within conducted experiments proven most successful in pellet bonding and limited compression, as a paste, consisting of brick debris (12%), earth-ware clay (18%), lime mortar (23%), and quartz sand (23%), and water (24%) formed by manual extrusion. The materials were selected for their representation of building materials and waste in the European Construction industry and their availability and abundance for studies within the scope of this research, and further providing large potential for scalability. This specifically applies to C&DW in Denmark, where structural mineral aggregates constitute at least 43% of all construction waste with a total of 2.185 kilotonnes in 2019 alone (Miljø- og Fødevareministeriet, 2020).

Through the aforementioned use of a closed-loop throwing system, precise material composition, local density and total weight are negligible. This is examined via a demonstrator (see figure 8).

Inter-robot communication

The implementation of the process relies on the intercommunication of all units of the robot ecology.





For continuous verification of the fabrication process each robotic unit functions in closed-loops exchanging data between the distributed units, transferred between units via TCP Client/Server connection.

While all data communication is tested between units a fully automated interpretation of the data at this time requires human aid and has not yet been tested without supervision.



Figure 8
Thrown and finished prototype

DISCUSSION: RESULTS AND OBSERVATIONS

The current progress of this research outlines the opportunities that a multi-robot RDM fabrication system can bring towards automation of on-site construction. At its current stage, the research development has focussed on the robot ecology and communications aspects. Working prototypes of each robot have been designed and fabricated, and the performance of each individual component to complete its tasks assessed through small case study scenarios.

The case study has also enabled testing an inter-robot communicative closed-loop approach. To allow scaling of the units and moving towards further automation of the process, all units should be integrated in ROS as shared platform. For full implementation as well as full automation of the process additional tests have to be conducted, and while communicative tests between the units have proven successful, at this stage of the research human supervision is needed, and a fully autonomous use of the proposed system could not yet be sufficiently tested.

The case study demonstrates that all three segments of the proposed robot ecology show a potential towards the automated use of materials with low embodied energy. This research proposes a material composition, allowing for reuse of C&DW material which fulfills expectations towards testing the robotic units, however lacks explorations towards full scale construction and evaluation over time.. While this research aims to propose a material to use, in order to evaluate use and feasibility of the developed robot ecology and further highlight the potential of reuse of CDW waste, within the current scope of this research variations of the material composition could not be sufficiently optimized. In the tests conducted, the aforementioned composition has proven to work best from samples conducted, but a higher ratio of C&DW would be desirable.

The next stage of research aims to scale up the testing, to more accurately assess the architectural scale(s) most suited to this approach. In this context,

the structural properties of the material, which can be characterized as a compressive material whose properties are largely driven by the ratios of the constituent materials, need to be more precisely assessed, and the consideration of energy and embedded carbon associated with robotic fabrication processes, which are potentially large, must be taken into account: Additional research towards life-cycle analysis should allow for further insight on the feasibility, sustainability and overall performance of the fabrication approach and material over time.

The research further shows the possibility of Additive Manufacturing specifically with untuned material. This is enabled based on the use of continuous closed -loop verification at all levels of the fabrication, in single unit operation as well as overall robot collaboration.

In summary, this research explores and demonstrates the potential of additive manufacturing with the use of construction and demolition waste via a distributed robot ecology, towards a more sustainably built future.

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